UNIT 2 · 8 hours

Time Domain Specifications



Time Domain Analysis of Control Systems

Control Systems (22EEG53)
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Learning objectives

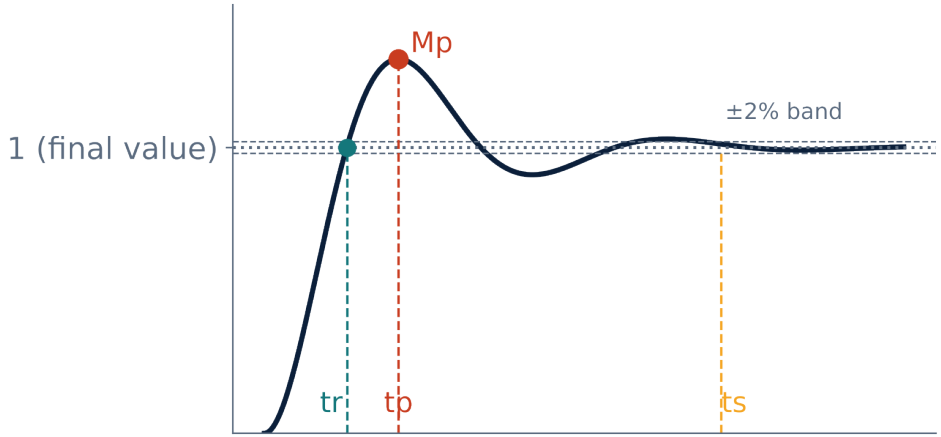
- Define delay, rise, peak, settling times and peak overshoot
- Derive the formulas for an underdamped second-order system
- Compute specifications from ωn and ζ
- Choose ωn and ζ to meet given specifications

Syllabus context

Time domain analysis of control systems: standard test signals, time response of zero, first and second order systems, time domain specifications, type of systems, static error constants and steady-state error, generalized error series.

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Transient specifications on the step response



- td: delay time (50 % of final value).
- tr: rise time (0 → 100 % for underdamped).
- tp: peak time.
- Mp: maximum (peak) overshoot.
- ts: settling time (±2 % or ±5 % band).

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Formulas (underdamped, $0 < \zeta < 1$)

$$t_r = (\pi - \varphi)/\omega_d$$

$\varphi = \cos^{-1}\zeta$

$$t_p = \pi/\omega_d$$

$\omega_d = \omega_n\sqrt{(1-\zeta^2)}$

$$M_p = e^{(-\pi\zeta/\sqrt{(1-\zeta^2)})} \times 100\ \%$$

overshoot

$$t_s = 4/(\zeta\omega_n) \quad (2\ \%)$$

= 3/(ζωn) for 5 %

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Observations

- Mp depends only on ζ: smaller ζ → larger overshoot.
- tr, tp, ts shrink as ωn increases for a given ζ.
- ts depends on the real part ζωn of the poles: move poles left → faster settling.
- Typical design range: ζ = 0.4–0.8 (Mp ≈ 25–1.5 %).
- Percent overshoot table: ζ = 0.2 → 52.7 %; 0.4 → 25.4 %; 0.6 → 9.5 %; 0.8 → 1.5 %.

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Numerical 1: compute specifications

Problem A system has $C/R = 25/(s^2 + 6s + 25)$. Find tr, tp, Mp and ts (2 %) for a unit step.

- $\omega_n = 5,\ \zeta = 0.6;\ \ \omega_d = 5\sqrt{(1 - 0.36)} = 4$ rad/s; $\ \varphi = \cos^{-1}0.6 = 0.927$ rad
- $t_r = (\pi - 0.927)/4 = 0.554$ s; $\ t_p = \pi/4 = 0.785$ s
- $M_p = \exp(-\pi \times 0.6/0.8) = e^{(-2.356)} = 9.48$ %
- $t_s = 4/(\zeta\omega_n) = 4/3 = 1.33$ s

Answer: tr = 0.554 s, tp = 0.785 s, Mp = 9.48 %, ts = 1.33 s

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Numerical 2: design to specifications

Problem Require Mp ≤ 10 % and ts ≤ 2 s (2 % band). Find the minimum ζ and ωn.

- Mp ≤ 10 %: $\zeta \geq \ln(10)/\sqrt{(\pi^2 + \ln^2(10))} = 2.303/3.895 = 0.591$
- $t_s = 4/(\zeta\omega_n) \leq 2 \Rightarrow \zeta\omega_n \geq 2$
- $\omega_n \geq 2/0.591 = 3.38$ rad/s
- Poles: $-2 \pm j2.73$

Answer: ζ ≥ 0.59, ωn ≥ 3.38 rad/s

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Numerical 3: servo

Problem Unity-feedback servo $G = K/[s(s+4)]$ with $K = 11.1$. Find ωn, ζ, Mp, ts.

- $s^2 + 4s + 11.1 = 0 \Rightarrow \omega n = 3.33,\ \zeta = 4/(2 \times 3.33) = 0.6$
- $Mp = 9.5\ \%$
- $ts = 4/(0.6 \times 3.33) = 2.0$ s; $\omega d = 2.67$; $tr = (\pi - 0.927)/2.67 = 0.83$ s

Answer: $\zeta = 0.6$, Mp = 9.5 %, ts = 2 s, tr = 0.83 s

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Key takeaways

- $t_r = (\pi - \varphi)/\omega_d$, $t_p = \pi/\omega_d$
- $M_p = \exp(-\pi\zeta/\sqrt{(1 - \zeta^2)})$; depends only on ζ
- $t_s = 4/(\zeta\omega_n)$ (2 %) or $3/(\zeta\omega_n)$ (5 %)
- Design: choose ζ for overshoot, ω_n for speed

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Practice questions

Q1. Derive the expressions for tp and Mp.

Q2. ωn = 10, ζ = 0.5: find tr, tp, Mp, ts.

Q3. Find ζ for Mp = 20 %.

Q4. Explain the effect of ωn on ts for fixed ζ.